

Depth-First Search Continued: Cycles, Bridges, and Articulation Nodes

Design and Analysis of Algorithms

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Cycle Detection

- Does an undirected graph have a cycle?
 - Note: If not, the graph would be a forest
 - YES, if a step of DFS takes you to a node you've seen before
- Does an directed graph have a directed cycle?
 - YES, if a step of DFS takes you to a “pending” node

Cycle Detection

- Suppose we modify DFS-Visit as follows:

```
DFS-Visit(i) :
```

```
  Status[i] = 'pending'
```

```
  For all j such that (i,j) is an edge:
```

```
    If Status[j] = 'pending': cycle detected!
```

```
    If Status[j] = 'unvisited': DFS-Visit(j)
```

```
  Status[i] = 'visited'
```

- Full-DFS will now detect the existence of a cycle in a directed graph in $O(m + n)$ time.
- In an undirected graph, the “m” term disappears!

Finding Bridges

- A **bridge** is an edge whose removal breaks a connected component into two pieces.
- An **articulation node** is a node whose removal breaks a connected component into two or more pieces.
- We can modify DFS to find all bridges and articulation nodes in a graph in $O(m + n)$ time.
- Let's focus on bridges for now (the approach for articulation nodes is quite similar).

Finding Bridges

DFS-Visit(i):

 Status[i] = 'visited'

$d[i]$ = current_time, Increment current_time

 min_d = $d[i]$

 For all $j \neq \text{Pred}[i]$ such that (i,j) is an edge:

 If Status[j] = 'unvisited':

 lowest_d_seen = DFS-Visit(j)

 If lowest_d_seen > $d[i]$ then (i,j) is a bridge!

 min_d = min(lowest_d_seen, min_d)

 Else, min_d = min($d[j]$, min_d)

 return min_d

